

Proof of conjecture 00000000960:
 $\ell^2(\mathbb{R})$ has no Schauder basis, but every separable subspace has one

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1 The conjecture and its reading

Statement (English, verbatim). Conjecture: There exists a Banach space with no Schauder basis, each of whose separable subspaces has one (the Enflo direction of the basis problem). The Chinese text says the same: there exists a Banach space with no Schauder basis, every separable subspace of which has a basis.

Reading. The claim is the existential statement

$\exists$ Banach space X : X has no Schauder basis $\wedge \forall V \leq X$ separable : V has a Schauder basis.

If such an X were separable, it would be a separable subspace of itself and would have a basis and also not have one. So any witness is non-separable. We prove the statement with $X = \ell^2(\mathbb{R}, \mathbb{K})$, $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Conventions (all used in the proof and the Lean code).

1. *Schauder basis* means the classical, sequence-indexed notion. A Schauder basis of a normed space Y is a sequence $(e_n)_{n \in \mathbb{N}}$ in Y with continuous linear functionals (f_n) such that $f_i(e_j) = \delta_{ij}$ and $\sum_{i < n} f_i(x)e_i \rightarrow x$ as $n \rightarrow \infty$ for every $x \in Y$. This is Mathlib's `SchauderBasis` $\mathbb{K} Y$ (an abbreviation of `GeneralSchauderBasis` $\mathbb{N} \mathbb{K} Y$ with the conditional summation filter), `Mathlib/Analysis/Normed/Module/Bases.lean`. Some authors also call an *uncountable* family a Schauder basis (Source 1 says so). Under that convention $\ell^2(\mathbb{R})$ has one, its orthonormal basis, and this proof does not apply. See Section 4.
2. *No basis* is proved in a strong form. X has no Mathlib `SchauderBasis`. More generally, X has no `GeneralSchauderBasis` indexed by any countable type along any non-trivial summation filter. It also has no sequence (e_n) such that every x is the limit of the partial sums of some series $\sum a_n e_n$. That last form needs neither uniqueness nor continuity of the coefficients, so it also rules out the definition in Source 1 (a sequence with unique expansions), with or without zero-padded finite bases.
3. *Separable subspace* means any linear subspace $V \leq X$ (closed or not) whose underlying set is separable, with the inherited norm. We do not restrict to closed subspaces.
4. *Finite-dimensional subspaces*. An $\mathbb{N}$ -indexed Schauder basis is linearly independent, so a finite-dimensional space never has one. For finite-dimensional V we give a finite basis: a `GeneralSchauderBasis` indexed by `Fin` $(\dim V)$ (biorthogonal, with $x = \sum_{i < \dim V} f_i(x)e_i$). For infinite-dimensional V we give a classical $\mathbb{N}$ -indexed Schauder basis.

The parenthetical. “The Enflo direction of the basis problem” refers to Banach’s basis problem. Source 1: “A Banach space with a Schauder basis is necessarily separable, but the converse is false. The basis problem is the question asked by Banach, whether every separable Banach space has a Schauder basis. This was negatively answered by Per Enflo who constructed a reflexive and separable Banach space failing the approximation property, thus a space without a Schauder basis.” (Enflo, Acta Math. 130 (1973) 309–317, <https://doi.org/10.1007/bf02392270>, Source 2.) Enflo’s space is separable, so it is a separable subspace of itself without a basis. It is therefore *not* a witness for the stated existential, which is a different and easier statement. We prove the statement as written.

2 Proof

Let $X = \ell^2(\mathbb{R}, \mathbb{K})$, the Hilbert space of square-summable families $(x_t)_{t \in \mathbb{R}}$. It is complete.

Lemma 1 (countable bases force separability). *Let Y be a normed space over $\mathbb{K}$ and $s \subseteq Y$ a countable set. If every $x \in Y$ lies in the closure of $\text{span } s$, then Y is separable. Hence a space with a Schauder basis, with a countably indexed generalized basis, or with a sequence (e_n) such that every x is a limit of partial sums $\sum_{i < n} a_i e_i$, is separable.*

Proof. The span of a countable set over the separable field $\mathbb{K}$ is separable, and so is its closure, which is Y . In each of the three cases every x is a limit of finite linear combinations of the countably many basis vectors. $\square$

Lemma 2. *X is not separable.*

Proof. Let $(b_t)_{t \in \mathbb{R}}$ be an orthonormal family indexed by $\mathbb{R}$ (Mathlib’s default Hilbert basis of $\ell^2(\mathbb{R}, \mathbb{K})$). For $s \neq t$, $\|b_s - b_t\|^2 = \|b_s\|^2 - 2\text{Re}\langle b_s, b_t \rangle + \|b_t\|^2 = 2$, so the open balls $B(b_t, 1/2)$ are pairwise disjoint. In a separable space a family of pairwise disjoint non-empty open sets is countable. But $\mathbb{R}$ is uncountable. $\square$

By Lemmas 1 and 2, X has no Schauder basis in any of the senses of Convention 2.

Lemma 3 (expansion in an incomplete space). *Let E be an inner product space, not necessarily complete, and let $(v_i)_{i \in I}$ be orthonormal with dense span. Then $x = \sum_i \langle v_i, x \rangle v_i$, with unconditional convergence, for every $x \in E$.*

Proof. Let $j : E \rightarrow \widehat{E}$ be the isometric embedding into the completion. The family (jv_i) is orthonormal, and its span $j(\text{span}\{v_i\})$ is dense, because j is continuous with dense range. So it is a Hilbert basis of $\widehat{E}$, and $jx = \sum_i \langle jv_i, jx \rangle jv_i = \sum_i \langle v_i, x \rangle jv_i$. Since j is a linear isometric embedding, the finite partial sums converge to x in E . $\square$

Lemma 4. *Let E be a separable inner product space, not necessarily complete. If E is infinite-dimensional, it has a classical Schauder basis. If it is finite-dimensional, it has a finite basis of length $\dim E$.*

Proof. Finite-dimensional case: take an orthonormal basis $(v_i)_{i < \dim E}$ and the coordinates $f_i = \langle v_i, \cdot \rangle$. Infinite-dimensional case: take a dense sequence (x_n) and apply the normalized Gram–Schmidt process. Discard the indices where the output is 0. This leaves an orthonormal family $(v_i)_{i \in S}$, $S \subseteq \mathbb{N}$, with $\text{span}\{v_i\} = \text{span}\{x_n\}$, which is dense. If S were finite, the expansion of Lemma 3 would be a finite sum and E would be finite-dimensional. So S is countably infinite. Enumerate it by $\mathbb{N}$, which gives an orthonormal sequence (e_n) with dense span. Let $f_n = \langle e_n, \cdot \rangle$.

These are continuous, and $f_m(e_n) = \delta_{mn}$. By Lemma 3, $\sum_n f_n(x)e_n$ converges unconditionally to x , so in particular its partial sums converge to x . $\square$

Theorem 5. $X = \ell^2(\mathbb{R}, \mathbb{K})$ is a Banach space without a Schauder basis. Every separable linear subspace $V \leq X$, closed or not, has a classical Schauder basis if it is infinite-dimensional and a finite basis if it is finite-dimensional. Moreover, X has closed, separable, infinite-dimensional subspaces, so the second clause is not vacuous.

Proof. The first claim is Lemmas 1 and 2. For the second, V with the inherited inner product is a separable inner product space, so Lemma 4 applies. For the third, the closure of $\text{span}\{b_n : n \in \mathbb{N}\}$ is closed and separable, and it contains infinitely many orthonormal, hence linearly independent, vectors. $\square$

3 Lean formalization and axiom audit

The file `lean/Conjecture960/Basic.lean` (256 lines, `import Mathlib` only) is byte-identical to the checked file. It uses the namespace `C960`. Below, K stands for the Lean scalar variable (a blackboard-bold k in the Lean source). It ranges over every `RLike` field in `Type`, so $\mathbb{R}$ or $\mathbb{C}$. X is `Mathlib's  $\ell^2(\mathbb{R}, \mathbb{K})$` , i.e. `lp (fun _ => K) 2` over the index type $\mathbb{R}$.

- `separableSpace_of_closure_span`, `separableSpace_of_series`, `separableSpace_of_generalSchauderBasis`: Lemma 1.
- `not_separableSpace_l2`: Lemma 2.
- `l2_no_countable_basis`: `IsEmpty (SchauderBasis K X)`, no countably indexed `GeneralSchauderBasis`, and no representing sequence.
- `hasSum_of_orthonormal_dense`, `onbBasis`: Lemma 3 and the resulting basis $(v_i, \langle v_i, \cdot \rangle)$. `toSchauder`: an $\mathbb{N}$ -indexed unconditional basis is a classical one.
- `exists_orthonormal_dense` (Gram–Schmidt, `Mathlib's gramSchmidtNormed`), `finiteDimensional_of_basis`, `finite_has_basis`, `separable_has_schauderBasis`: Lemma 4.
- `l2_separable_subspace_has_basis`, `l2_exists_closed_separable_infinite`.
- The main theorem `conjecture960` states that there exist $X : \text{Type}$, `NormedAddCommGroup X` and `NormedSpace K X` such that:
 - `CompleteSpace X` and `not (SeparableSpace X)`;
 - `IsEmpty (SchauderBasis K X)`;
 - for every countable `iota : Type` and every `SummationFilter L` on it with `L.NeBot`, `IsEmpty (GeneralSchauderBasis iota K X L)`;
 - there is no $e : \mathbb{N} \rightarrow X$ such that every x has scalars $a : \mathbb{N} \rightarrow \mathbb{K}$ with $\sum_{i < n} a_i e_i \rightarrow x$ (`Tendsto ... atTop (nhds x)`);
 - for every $V : \text{Submodule } K X$ with `IsSeparable (V : Set X)`, `FiniteDimensional K V` implies `Nonempty (UnconditionalSchauderBasis (Fin (Module.finrank K V)) K V)`, and `not (FiniteDimensional K V)` implies `Nonempty (SchauderBasis K V)`;
 - some $V : \text{Submodule } K X$ is closed, separable and not finite-dimensional.

- `conjecture960_real` restates the real case in the form closest to the wording: a complete real normed space with `IsEmpty (SchauderBasis R X)` in which every separable, infinite-dimensional subspace has a `SchauderBasis`.

Axiom audit. `#print axioms` for `C960.conjecture960` and `C960.conjecture960_real` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom. `lake build` succeeds (Lean v4.33.1, Mathlib v4.33.1).

4 What is not covered

Under the convention that admits uncountable, linearly ordered “Schauder bases” (Source 1: “some authors define Schauder bases to be countable (as above), while others use the term to include uncountable bases”), $\ell^2(\mathbb{R})$ has a basis, its orthonormal basis, and is not a witness. We do not address that reading. The conjecture says “Schauder basis”, and the definition quoted in Source 1 (“A Schauder basis is a sequence $\{b_n\}$...”) is sequence-indexed, so we use the countable notion.

Sources

1. Wikipedia, “Schauder basis”, https://en.wikipedia.org/wiki/Schauder_basis (retrieved 2026-10-05). Quotes used: “A Schauder basis is a sequence $\{b_n\}$ of elements of V such that for every element $v \in V$ there exists a unique sequence $\{\alpha_n\}$ of scalars in F so that $v = \sum \alpha_n b_n$ ”; “Note that some authors define Schauder bases to be countable (as above), while others use the term to include uncountable bases”; and the basis-problem passage quoted in Section 1.
2. P. Enflo, “A counterexample to the approximation problem in Banach spaces”, *Acta Mathematica* 130 (1973) 309–317, <https://doi.org/10.1007/bf02392270> (Crossref record retrieved 2026-10-05).
3. Mathlib: `Mathlib/Analysis/Normed/Module/Bases.lean` (`SchauderBasis`, `GeneralSchauderBasis`), `Mathlib/Analysis/InnerProductSpace/l2Space.lean` (`HilbertBasis.mk`, `hasSum_repr`), `Mathlib/Analysis/InnerProductSpace/GramSchmidtOrtho.lean`.